Session 6: The Internet

Tech Skills 101: Driver's Ed for the Digital World

David Shamlin
OLLI Fall 2025

Agenda

- What the Internet and Web are
- How browsers interact with sites on the Internet
- What a URL is and how to “read” them
- Time permitting
    - Cookies
    - Private browsing
    - The Dark Web

Terminology

site	A site (a.k.a., “website” is a presence on the World Wide Web. As of 2025, there are more than one billion sites. Sites are identified to the public by their "domain name," such as “apple.com" or "pcmag.com".
page	A page (a.k.a., “web page”) is one of many files that make up a website. A page is actually a text file coded in HTML and JavaScript with links to separate files (images, videos, buttons, etc.) that also appear on the page.
URL	Essentially, the address of a page. A URL includes the name of a site and the name of a page at that site that you want to view.
link	A link (a.k.a. “hyperlink”) typically refers to an icon or text on a page that, when clicked or tapped, transfers the user to another part of the page or to another page on the current website or to a completely different website.
browser	A browser (a.k.a., “web browser”) is an app for accessing sites. The purpose of a browser is to fetch content from a site and display it on the your device. When a you request a page from a particular website, the browser retrieves its files from a web server and then displays the page on the your screen.
address bar	An address bar is a text field near the top of a browser window that displays the URL of the current page. The URL, reflects the address of the current page and automatically changes whenever you visit a new page. Therefore, you can always check the location of the page you are currently viewing with the browser's address bar. While the URL in the address bar updates automatically when you visit a new page, you can also manually enter a web address. Therefore, if you know the URL of a site or specific page you want to visit, you can type the URL in the address bar and press Enter to open the location in your browser.
HTTP / HTTPS	An acronym for “Hypertext Transport Protocol”, HTTP is the software apps use to communicate with sites. HTTP is software itself; a copy of HTTP is included with operating systems. The “S” in HTTPS stands for “secure” and means that the communication between your app and a site is encrypted.
IP address	A number that uniquely identifies your device when accessing the Internet. A devices’s IP address may be permanently assigned or supplied each time that it connects to the Internet by an Internet service provider. There are two different kinds of IP Addresses being used today: IPv4 and IPv6; IPv4 is being “phased out” in favor of IPv6 due to the exponential growth of the Internet.
cookie	*See slide titled “Definition of Cookie”.*

Terminology: the *Internet* vs *Web*

The ***internet*** is a global network of billions of servers, computers, and other hardware devices. Each device can connect with any other device as long as both are connected to the internet using a valid IP address. The internet makes the information sharing system known as the web possible.

The ***web*** (a.k.a., the "World Wide Web") is one of the ways information is shared on the internet (others include email, File Transfer Protocol (FTP), and instant messaging services). The web is composed of billions of connected digital documents that are viewed in a web browser, such as Chrome, Safari, Microsoft Edge, Firefox, and others.

Think of the internet as a library. Think of the books, magazines, newspapers, DVDs, audiobooks, and other media it contains as websites.

Both the internet and the web serve unique purposes but work hand in hand to provide information, entertainment, and other services to the public.

https://olli-shamlin.github.io/spring-2024/syllabus.html

Uniform Resource Locator

Protocol

Resource

https://olli-shamlin.github.io/spring-2024/syllabus.html

Host Name
or Domain Name

Anatomy of a URL

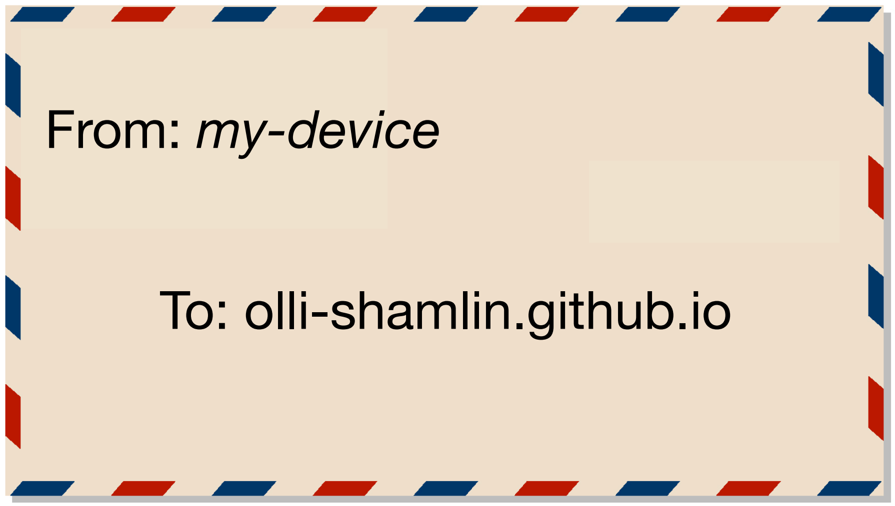
*protocol***://***hostname***/***resource***?***parameters*

Example of a URL with parameters

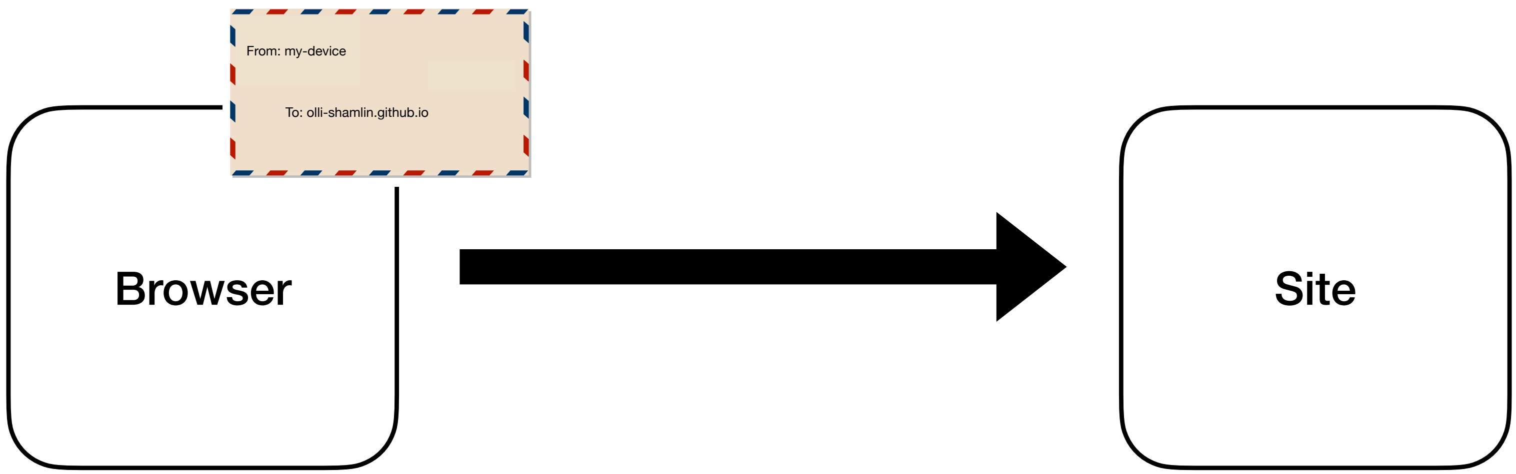
https://www.google.com/search?
q=what+is+a+url&sca_esv=ef32433310457e91&sca_upv=1&ei=FUwiZsuiFcOs5No
P_7CMyAk&ved=0ahUKEwiLgcOrjc6FAxVDFlkFHX8YA5kQ4dUDCBA&uact=5&oq
=what+is+a+url&gs_lp=Egxnd3Mtd2l6LXNlcnAiDXdoYXQgaXMgYSB1cmwyCBAA
GIAEGLEDMgUQABiABDIFEAAYgAQyBRAAGIAEMgUQABiABDIFEAAYgAQyBRA
AGIAEMgUQABiABDIFEAAYgAQyBRAAGIAESPAeUMYNWJgccAR4AZABAJgBZK
ABuAaqAQQxMi4xuAEDyAEA-
AEBmAIRoAKOB8ICChAAGLADGNYEGEfCAg0QABiABBiwAxhDGIoFwgITEC4Yg
AQYsAMYQxjIAxiKBdgBAcICEBAAGIAEGLEDGEMYgwEYigXCAgsQLhiABBixAxi
DAcICCxAAGIAEGLEDGIMBwgIREC4YgAQYsQMY0QMYgwEYxwHCAg4QLhiAB
BixAxjRAxjHAcICChAAGIAEGEMYigXCAg4QLhiABBixAxiDARiKBcICDhAAGIAEGL
EDGIMBGIoFwgINEAAYgAQYsQMYQxiKBcICCBAAGIAEGMkDwgILEAAYgAQYkg
MYigXCAgsQLhiABBiRAhiKBcICBBAAGAPCAiMQLhiABBiRAhiKBRiXBRjcBBjeBB
jgBBj0AxjxAxj1A9gBApgDAIgGAZAGDLoGBAgBGAi6BgYIAhABGBSSBwQxNi4xo
AeVZg&sclient=gws-wiz-serp

https://olli-shamlin.github.io/spring-2024/syllabus.html

https://olli-shamlin.github.io/spring-2024/syllabus.html



https://olli-shamlin.github.io/spring-2024/syllabus.html

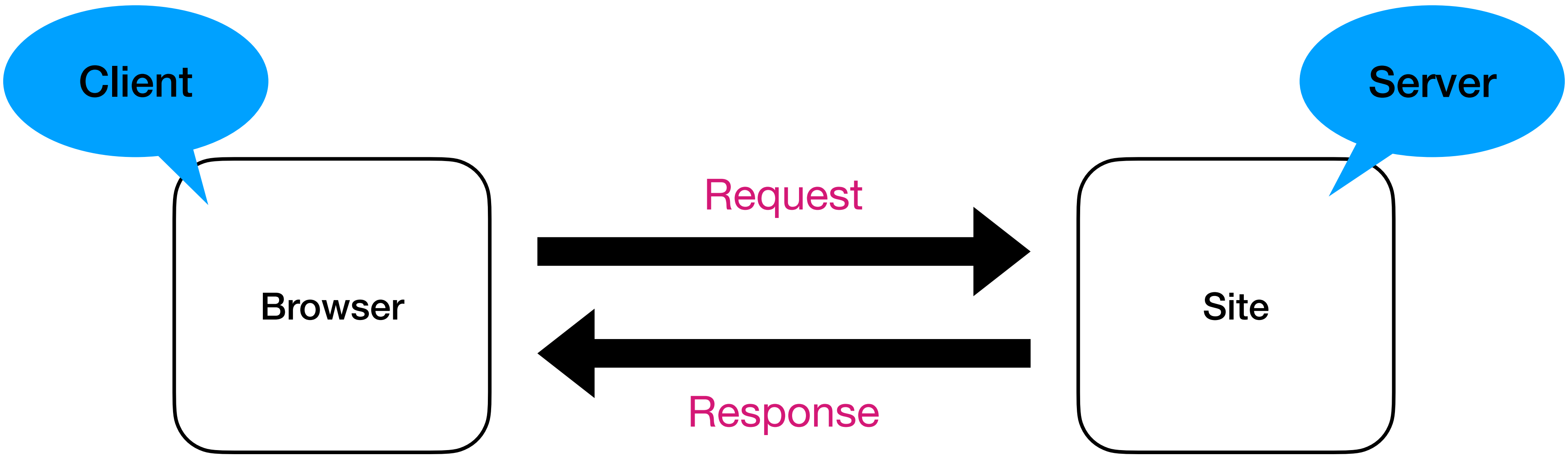


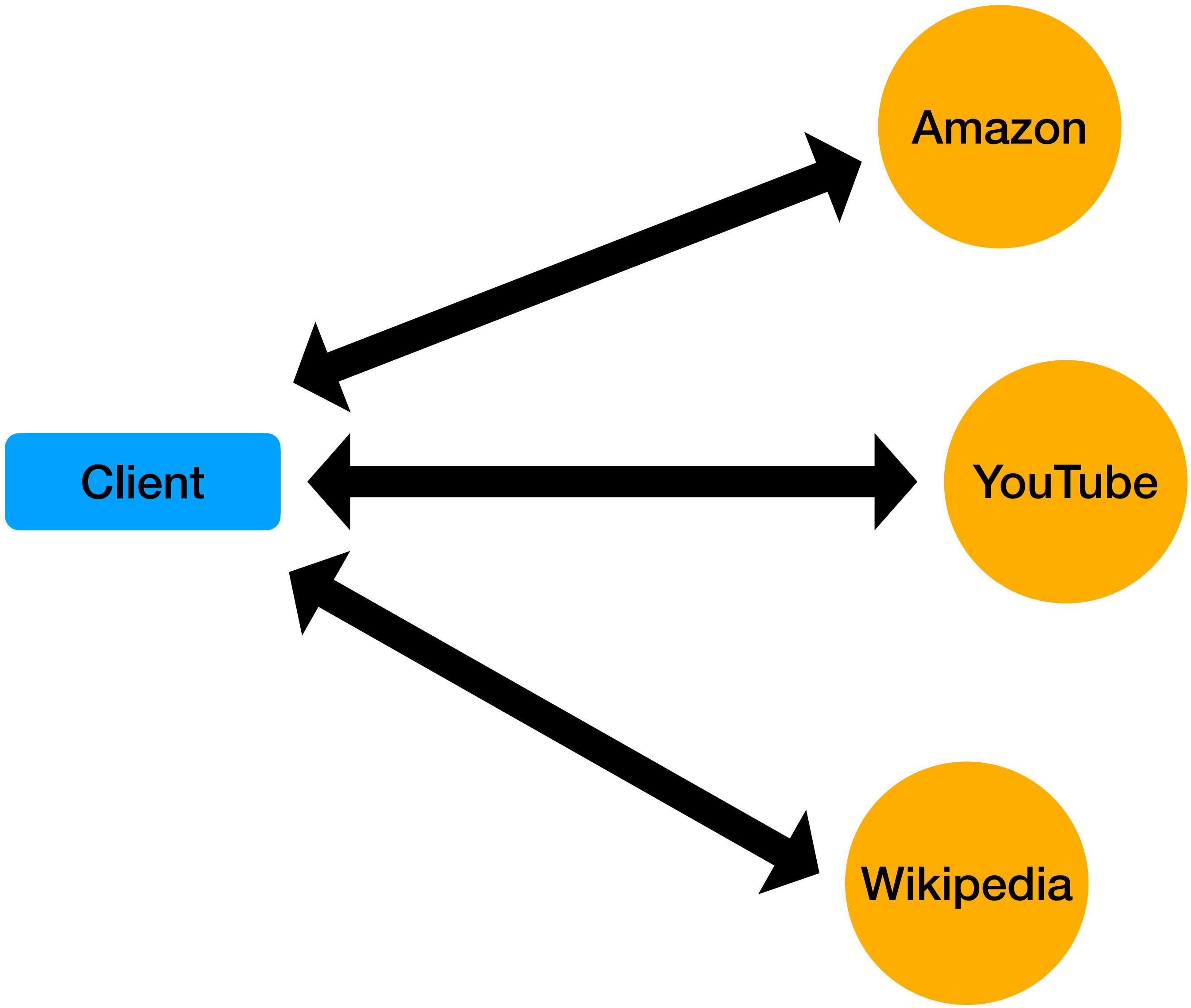

<h2>Course Aims and Outcomes</h2>
<div class="container">
<p>
This course is designed to increase your digital proficiency specifically as it perta
Internet. Lectures presented over the course's sessions will give you a working un
ow the Internet, the World Wide Web, and the Cloud work. While lectures will incl
ussions of how computers and software used with the Internet work, the lectu
s a general audience. There is no prerequisite technical understanding need
ticipate in this course.


HTML
Hyper Text Markup Language

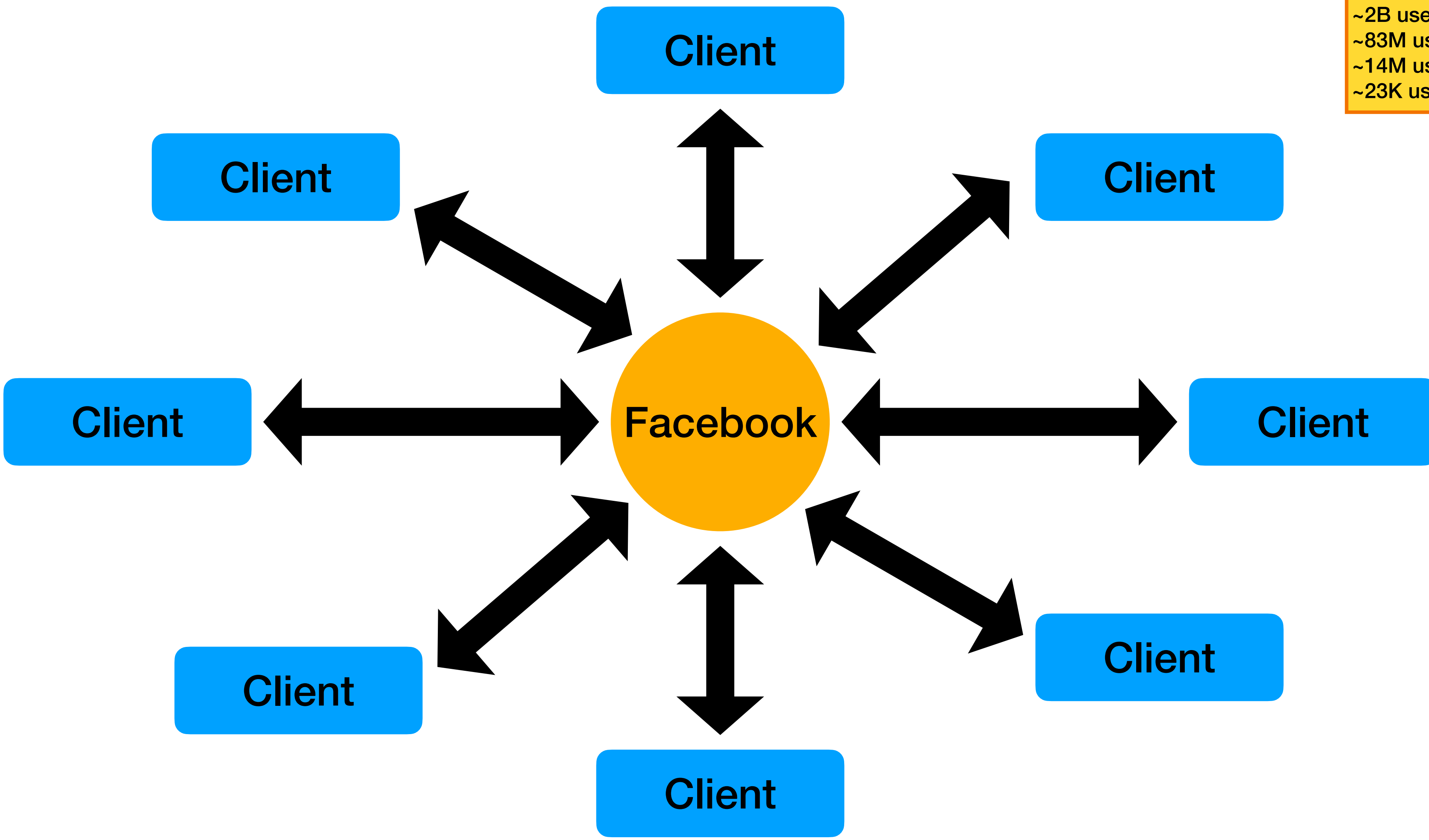

From: olli-shamlin.github.io
To: *my-device*


Browser
Site
From: olli-shamlin.github.io
To: *my-device*

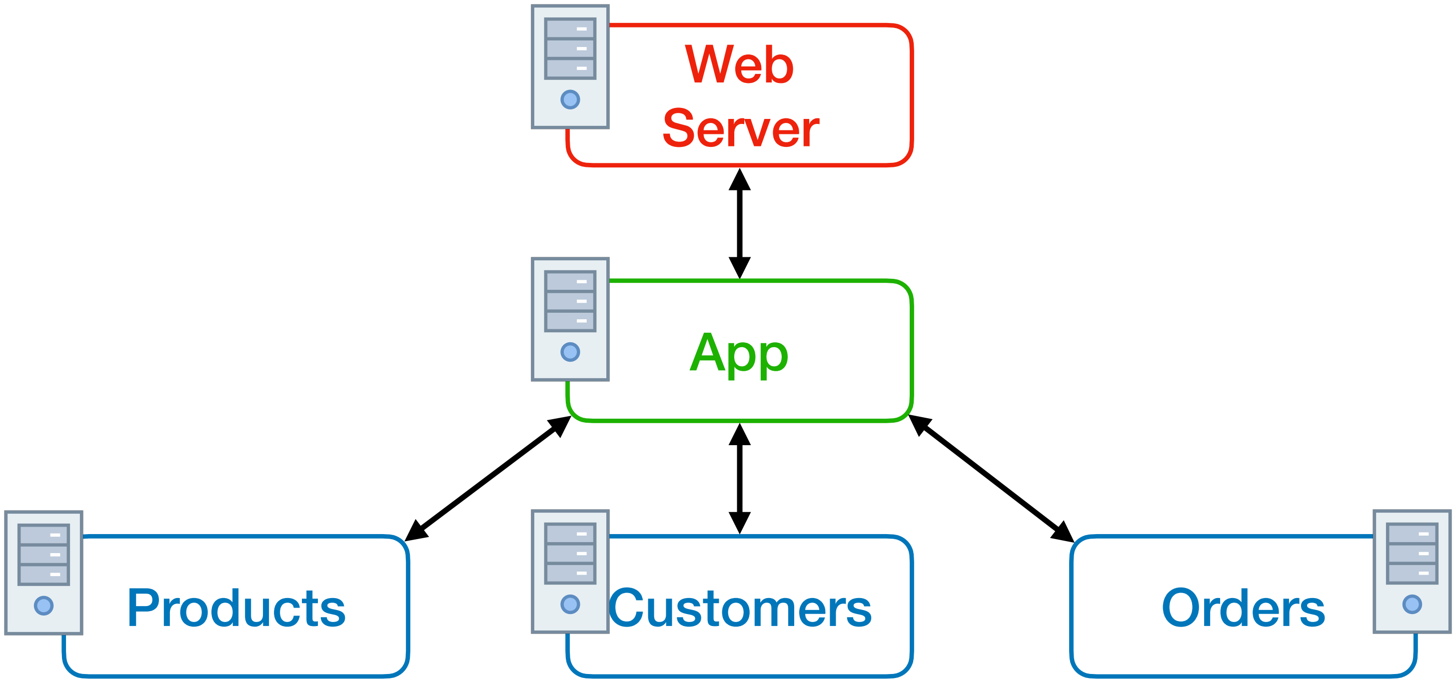

Client
Server
Browser
Request
Response
Site


Amazon
Client
YouTube
Wikipedia

Facebook
~2B users/day
~83M users/hour
~14M users/min
~23K users/sec

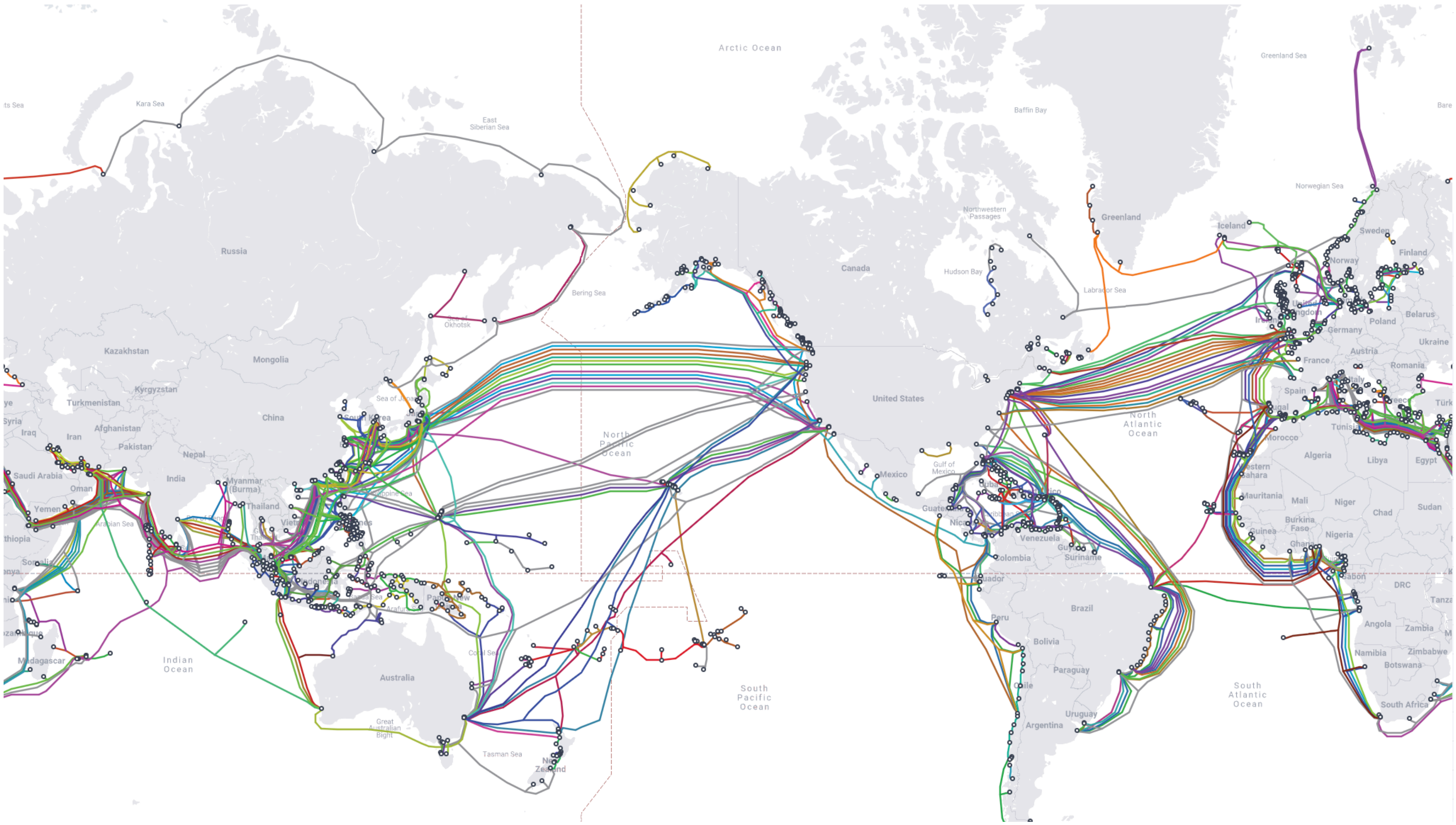


Example: Retail Site

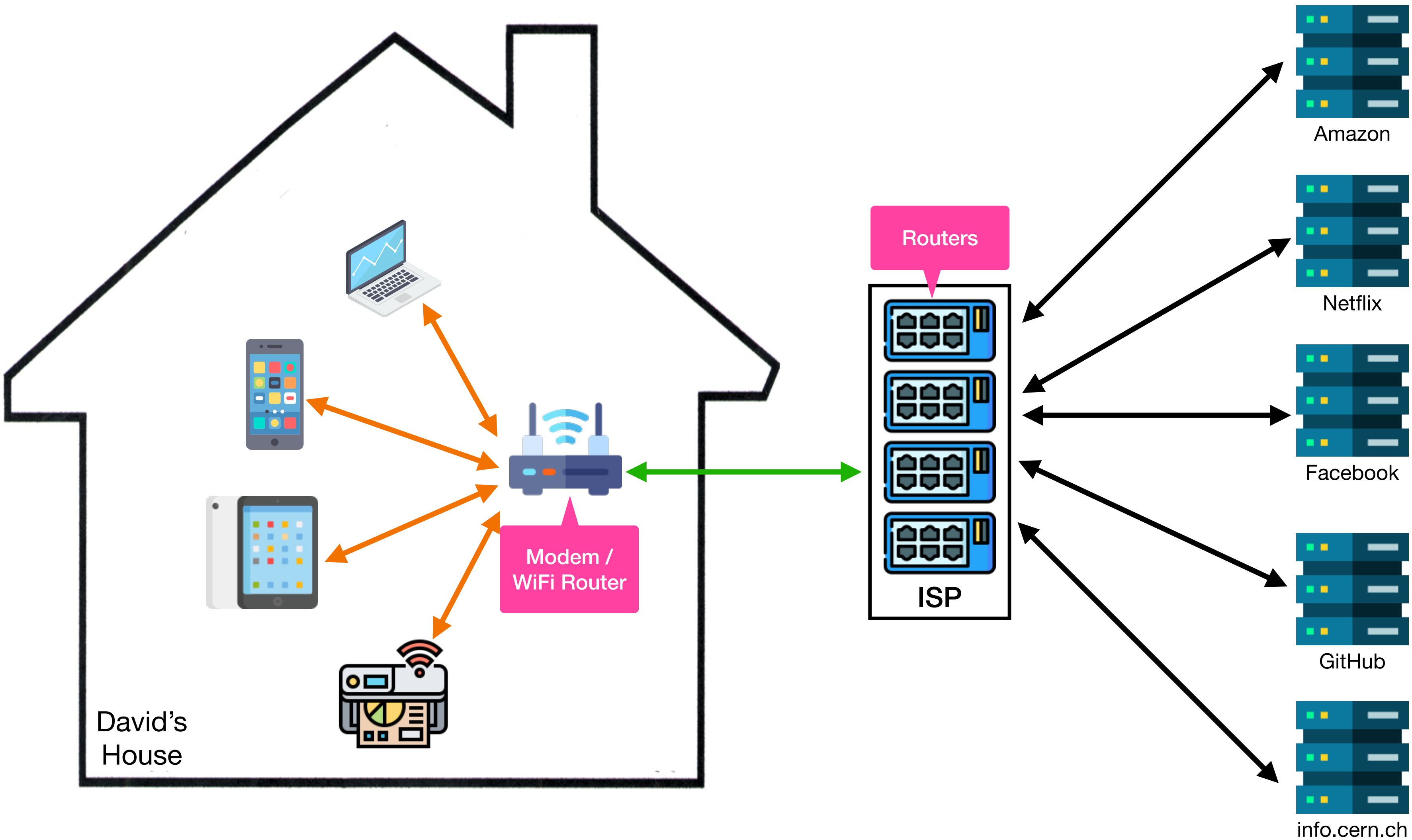



14
13
12
11
10
9
Power
Speed
Link/Act

The Internet Backbone

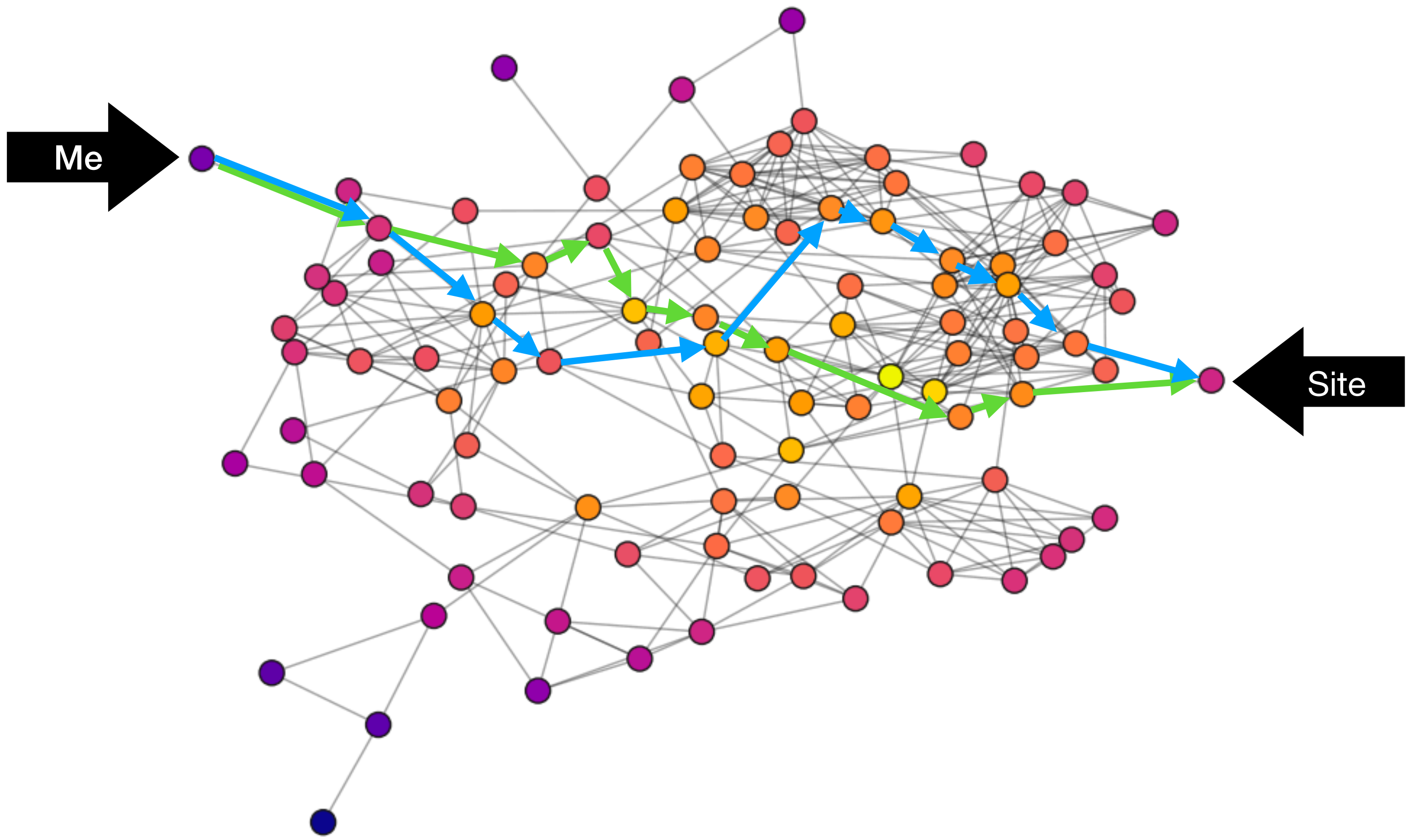


The Internet Backbone


Amazon
Netflix
Facebook
GitHub
info.cern.ch
Routers
ISP
Modem /
WiFi Router
David's
House

Routing

How packets find their way from point A to point B on the Internet



Definition of *Cookie*

From wikipedia.org:

An HTTP ***cookie*** (also called web cookie, Internet cookie, browser cookie, or simply cookie) is a small block of data created by a web server while a user is browsing a website and placed on the user's computer or other device by the user's web browser. Cookies are placed on the device used to access a website, and more than one cookie may be placed on a user's device during a session.

Cookies serve useful and sometimes essential functions on the web. They enable web servers to store stateful information (such as items added in the shopping cart in an online store) on the user's device or to track the user's browsing activity (including clicking particular buttons, logging in, or recording which pages were visited in the past). They can also be used to save information that the user previously entered into form fields, such as names, addresses, passwords, and payment card numbers for subsequent use.

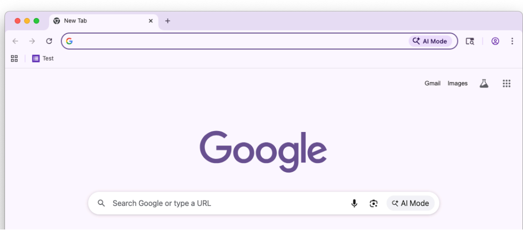
Examples:

av-timezone	America/New_York
csm-hit	tb:s-M6XQHR0B3T548ZJE02PH|1760965558625&t:1760965559766&adb:adblk_no
i18n-prefs	USD
lc-main	en_US
session-token	b+iYdts4XRFV5TD7RsVUPyK6ZQtJZDLsUkkagDgoTJNVwdBKtFIgOaMCwy33KB122EYcaFUSt1oGX

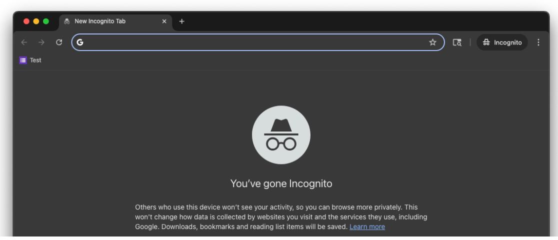
Private Browsing Mode

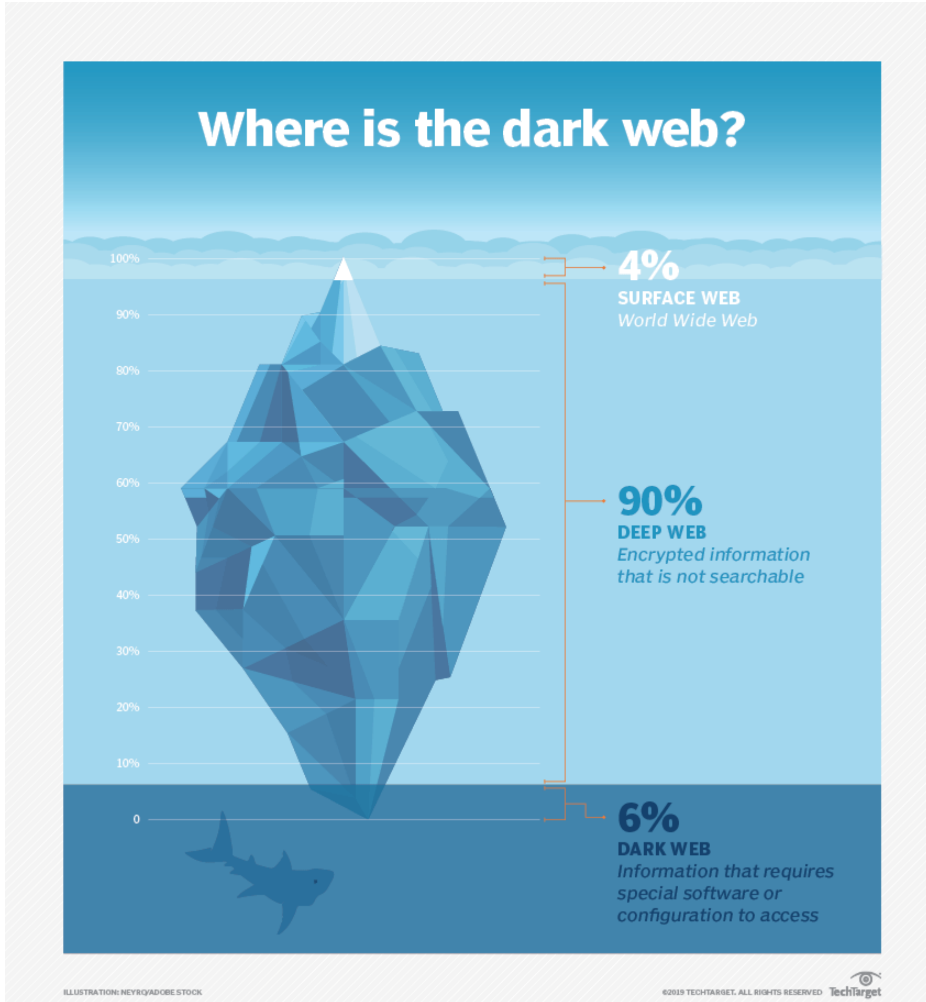
- Browsers save some info when we use them
    - Cookies
    - History of sites/pages you visit and search queries you make
    - Large files (a.k.a. “cached files”; eg, images and videos)
- You can use a browser’s “private browsing” feature to prevent your browser from saving the above types of data
- Private browsing mode inhibits others who you share your device with from seeing your browsing activity; it **does not** increase your privacy to others on the internet who may be “watching” (e.g., the sites you visit, your ISP, “bad guys” trying to scam or hack you)
- **Note:** Google Chrome calls this “incognito mode” / “incognito windows”

Regular window



Private window




Where is the dark web?
100%
90%
80%
70%
60%
50%
40%
30%
20%
10%
0
4%
SURFACE WEB
World Wide Web
90%
DEEP WEB
Encrypted information that is not searchable
6%
DARK WEB
Information that requires special software or configuration to access
ILLUSTRATION: NEYRO/ADOBE STOCK
©2019 TECHTARGET. ALL RIGHTS RESERVED
TechTarget

cit-666